



Utilization of ERP systems in manufacturing industry for productivity improvement

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ABSTRACT

In commercial enterprises, the Enterprise Resource Planning (ERP) system is a well-known and frequently used solution for process integration and automation, performance enhancements, and cost savings. Enterprise Resource Planning (ERP) systems are practical information technology (IT) solutions that provide managers with sufficient data in research. ERP collects data from all of a company's functions, giving the entire organisation a broader perspective. ERP systems are capable of controlling the entire firm by tracking supply, requests, scheduling, finished inventory goods, and other critical information that is necessary to management. By adding functionality to previous industrial resource planning systems, the purpose is to harmonise operations and procedures inside a manufacturing organisation. Successfully deployed Enterprise Resource Planning systems can deliver considerable strategic, operational, and information-related benefits to the relevant enterprises. The majority of data on the failures and achievements of such systems are now based on reports from large manufacturing and service firms. ERP vendors, on the other hand, are increasingly focusing their marketing efforts on small and medium-sized organisations. We address some of the implications of ERP systems for the next generation of prospective departmental applications in Manufacturing and Enterprise Management in this article.

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1. Introduction

Our manufacturing industry has shifted over time, and the speed of development has intensified as global companies try to fulfil new customer demands and grow into new areas. Successful firms recognise the importance of integrated systems in improving quality, customer satisfaction, performance, and profit. The effective transmission of data in an enterprise is crucial from the aspect of business management. The company cannot function efficiently unless it has accurate, trustworthy, and timely data. As a result, appropriate management information systems must be put in place. Organizations can accomplish this objective by implementing ERP solution technologies [26] ERP is a sort of corporate management software that merges a company's information and procedures into a single system. It has a software architecture that encourages data flow across all business functions. Furthermore, it facilitates data and information transfer, allowing for the automation and integration of corporate processes in order to attain best

practices in process management [1,2]. One of the primary purposes of implementing ERP is to link corporate activities, which is critical in enhancing customer satisfaction and decreasing manufacturing wait times. Other potential ERP advantages involve considerable significant capital investment savings, a wealth of knowledge about consumer wants and wishes, and the ability to view and govern the entire enterprise of vendors, partnerships, and clients as a form of discipline. It is considered as an integrated, customized, packaged software-based system that manages most of an enterprise's software systems across all functional areas, including automation, finance, human resources, supply chain, manufacturing, sales, and marketing. ERP systems provide quicker and more accurate control of a business's changes, which is especially vital in an era of inescapable competition and constant economic and legal changes [27]. Therefore, it can be assumed that the implementation of ERP systems can result in significant benefits and a relatively quick return on investment. In this paper, we present the adoption of enterprise resource planning (ERP) systems in various kinds of business organizations and organize a literature review on their success. Next, a methodology is provided to demonstrate the process of its implementation. Following that,

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some applications are explored, demonstrating how ERP is deployed in those industries and how it affects corporate performance. A reference to the results obtained from a survey from Ranjan, S., Jha, V.K., and Pal, has been considered as it leads to understanding efficiency projections across multiple factors of commonalities, provided implementation and pre-design planning are done considering all requirements of the organization efficiently [28]. The final section of the paper summarizes the process and makes recommendations for further research [3].

2. Literature review

The interrelationship between ERP systems and their application is a relatively new area of study that is still evolving. ERP systems are database systems that obtain data to help manage medium and large organizations. ERP systems marked the start of a new age in business, allowing firms to connect business processes and respond to real-time data. Despite this, the relevant research has primarily focused on ERP systems in general, with little published scientific data on ERP integration activities and their implications on an application in particular. ERP systems assist businesses by tightly connecting different departments [25]. According to recent studies, ERP system adoption has an impact on financial accounting, management accounting, manufacturing management, automation, and supply chain management. ERP systems appear to be a driving force behind the adoption of modern industrial techniques, with ERP adopters and non-adopters differing. It also shows that ERP systems minimize the amount of time required to complete mundane activities, giving employees more opportunity to undertake more meaningful production analyses [3]. Without the assistance of an effective IT system, today's large industrial businesses are unable to function efficiently. The digitalization of all aspects of a company's operations boosts productivity improvement and effectiveness, as well as its market competitiveness. A multi-module ERP system is required, rather than basic software. Because an ERP system-using firm is reliant on the system, it's critical to tailor it to the company's needs early in the installation process. Enterprises considering implementing an ERP system should pay special attention to selecting a specific, dependable system solution that suits their needs. The study conducted in this article will specifically evaluate the influence of ERP systems on their applications [32]. The main properties of ERP systems are discussed in this article, as well as the application areas in a business. ERP systems are vast and complicated, and their deployment requires careful planning and execution. They aren't just software systems; they have an impact on how a company operates. Whether an ERP system provides a competitive advantage or becomes a corporate burden is determined by how it is implemented. Because ERP implementations require major changes to established business procedures as well as large capital expenditures, senior management engagement is the most important factor in ensuring a successful ERP installation. The challenges of reengineering business processes and connecting other business applications with the ERP backbone are also key considerations. The benefits and relevance of putting these systems in place have been given special consideration [4]. One of the quickest and most successful methods to achieve corporate transformation is to implement new Information Systems (IS), such as an Enterprise Resource Planning (ERP) system. ERP deployment is critical for firms to develop a solid IT infrastructure that unifies business operations and automates and integrates fundamental processes across departments. The selection or design of an ERP system is the primary issue that every business has while transitioning to it. This factor has a significant impact on an organization's operations, pro-

cesses, and activities both during the implementation phase and thereafter in the administration and maintenance process [31].

3. Problem statement

Small and Medium Enterprises (SME) may find it challenging to report and analyse business progress in real-time without an ERP system. By investing in such a system, managers will be able to view and export reports based on the most up-to-date business progress within the organization. ERP solutions consolidate all costs, marketing, and sales data into one system, ensuring that reports are accurate and updated and also manage modern manufacturing processes involving automation and robotics.

4. Methodology

Most businesses have learned that in today's fast-paced world, it's hard to build and manage a bespoke software package that meets all of their needs while also remaining relevant. Recognizing customer demands, several of the world's most well-known software companies have developed Enterprise Resource Planning software, which will give an integrated software solution for all of an organization's activities. By developing and maintaining a single database of corporate data, many organizations that utilize ERP systems want to reduce data redundancy and inconsistency [4].

4.1. Features of ERP

- Manufacturing, Selling and Distribution, Inventory, Payables, Receivables, Accounts, Purchases, Human Resources, and so on are all covered by ERP,
- ERP automates essential business functions and enhances customer service while also boosting the company's image and bridging the information gap.
- ERP allows for the seamless integration of cutting-edge technologies like Electronic Data Interchange (EDI), Electronic Fund Transfer (EFT), Internet, and E-Commerce.
- ERP addresses material shortages, productivity improvements, quality issues, customer service, financial management, and prompt delivery.
- Business intelligence technologies such as Decision Support Systems (DSS), Reporting, Data Mining, Executive Information Systems (EIS), and Early Warning Systems (Robots) are offered in ERP to help employees make better decisions and enhance their business operations [6].

4.2. Implementing ERP

When a firm needs specific ERP software, it must make budgetary sacrifices since redesigning modules and delivering ERP software is an expensive endeavour. This is due to the difficulties that come along with it. It necessitates the use of people with more practical experience to develop the systems. It will also be necessary to update the procedure regularly. Given the numerous modifications made to the system to make it more business-friendly, this will be challenging. The selection of the appropriate ERP solution is a critical stage. The following steps are involved in the evaluation and selection process:

- Determining whether all functional aspects of the business are adequately covered.
- Determining whether all business operations and activities are fully integrated.

- Determining whether the vendor has personalizing and implementation capabilities.
- Determining whether all current IT trends are addressed.
- Determining whether the business can absorb the cost and achieve the best return on investment.

The implementation of an ERP system must be done in stages. The overall time needed to effectively integrate an ERP software will range from 18 to 24 months. The following are the typical steps involved in ERP implementation: Gap analysis as a part of the operational and business analysis, Project Scheduling, Training for the project team, Restructuring of Business Processes, Configuration and installation, Mapping of Business Requirements, Conformity testing, and Assistance with post-audits/implementation [5,30].

4.3. Benefits of ERP

See Table 1.

4.4. Risks involved

- High initial investments - ERP systems are expensive because they need high initial infrastructure expenditures. A solid IT system, a decent networking system, and a robust reporting system are required for these systems. A sloppy IT infrastructure might lead to issues like the ERP system's sluggish processing speed. Due to the complexity of ERP systems, organizations must collaborate with a team of experts.
- Lack of compatibility with ERP software - There is a chance that the ERP software you choose isn't a suitable fit for your company's needs. This might happen because of a bad review by experts that are assisting the company in finding the finest ERP software.
- Team members have a high rate of turnover - Due to the heavy workload and high levels of job stress, coordination is put under tremendous strain, resulting in a high rate of turnover. Due to a short timeline, a tight deadline to finish, a lack of team members' experience, and management inflexibility, this happened in several organizations.
- The commitment of the senior management - According to IT literature, senior management support is required for IT projects to succeed. When it comes to adopting an ERP system, it's not only about changing software systems; it's also about repositioning the organization and altering business processes [6].

Table 1
Five aspects of ERP advantages are defined.

No.	ERP Benefits	Definition
1	Operational benefits	The operational benefits are those arising from automating cross functional processes.
2	Managerial benefits	This includes benefits that arise from the use of data to better plan and manage production, manpower, inventory and physical resources and from the monitoring and control of financial performance of products, customers, business lines and geographic area.
3	Strategic benefits	This category focuses on the benefits that arise from the system's ability to support business growth.
4	IT infrastructure benefits	This consists of the typical IT department benefits arising from reduction in cost of maintaining legacy systems.
5	Organizational benefits	This category captures the benefits derived from facilitation business learning, empowerment of staff and higher employee morale and satisfaction.

4.5. Research review analysis

Ranjan, S., Jha, V.K., and Pal, Surveyed with 25 questions, one for each benefit element, to rate their importance in light of the primary notion, namely, the benefits of ERP latest technological implementation. Seven questions for “**employee**,” seven questions for “**operation**,” six questions for “**client**,” and five questions for “**financing**” were divided into four subsets that addressed all objective indicator areas. Each of these subgroups focused on a different aspect of new technology, such as social networking, big data analytics, mobile computing, Internet of Things and cloud computing (Table 2).

Few essential fundamental criteria were observed when framing the questions, such as each question's relationship to the study's goal was clearly defined, and ambiguities were avoided. The questionnaire's online link was initially sent to 200 people, and the first batch yielded about 150 replies. Follow-up emails were sent, as well as another 200 circulations, from which about 100 answers were gathered. The overall response indicated 63% of effectiveness provided by ERP and new technologies towards efficiency in Indian manufacturing enterprises. They evaluated their findings in a recent study. While the majority of studies focused on the advantages of ERP and new technologies, some also include risk concerns related to inefficient pre-planning by developers during system design stages. It is critical to emphasize objective organizational indicators and strategic IT alignment for organizational transformation when identifying the benefit elements.

The research yielded eight major benefit elements that address all four objective metrics. The following are the advantages that are prioritized in descending order of importance for each objective indicator:

1. With transformation driver “learning maturity,” two elements pertaining to “employee” with overall greatest importance allocated “Knowledge management and IP protection.”
2. Two variables in the target indicator “client” with the greatest importance are associated with “capturing individual user experiences,” while the transformation driver is “product customization and integration.”
3. Three variables from the target indicator “operation” with the greatest importance are associated with “faster NPI and increased efficiency,” and the transformation driver is “production competence.”
4. “Decisive financial decision making skill” is one component in objective indicator “financing” with priority; transformation driver is “financial assertiveness” [28].

5. Importance of ERP in manufacturing management

5.1. Manufacturing management

ERP is the most recent stage in the evolution and expansion of production planning and control methodologies for manufacturing companies, ranging from material requirements planning (MRP) to capacity needs planning (CRP) to Manufacturing Resource Planning (MRP II). ERP systems are made up of a collection of software modules, each of which is responsible for gathering and processing data for a certain business function or group of related business functions. Accounting, master scheduling, material planning, inventory, forecasting, finite scheduling, distribution planning, and other ERP software modules are also commonly used. Furthermore, we will examine how some of these processes are managed by software modules when they are used in Manufacturing Operations and Sectors [9].

Table 2
Communalities of factors.

No.	Components	Initial	Extraction	No.	Components	Initial	Extraction
1	Talent acquisition and retention	1.000	0.686	13	Low inventory and supply chain alignment	1.000	0.636
2	Employees that are productive and committed	1.000	0.830	14	Agile computing, Reliable, scalable	1.000	0.662
3	Employees with more technical maturity	1.000	0.718	15	Individual user experiences are being gathered	1.000	0.742
4	Employees that are project oriented	1.000	0.768	16	Co-creation and collaboration on a large scale	1.000	0.779
5	Decisive insights in data ocean	1.000	0.728	17	Clients and retailers' brand loyalty	1.000	0.716
6	Improve the availability of employees	1.000	0.528	18	IoT is used to promote consumerism	1.000	0.602
7	Knowledge management and IP protection	1.000	0.860	19	Easier life-cycle support and disposal	1.000	0.564
8	Sensing the pulse of stakeholders	1.000	0.788	20	Voice of customers and pulse of market	1.000	0.767
9	Low security threat to vital business data	1.000	0.748	21	Impact on organizational value chain	1.000	0.769
10	More opportunities for creative thinking	1.000	0.781	22	Strategic financial decision. making ability	1.000	0.749
11	Refinement of business processes and interfaces	1.000	0.600	23	Low carbon footprint due to cloud ERP	1.000	0.690
12	Superior product quality and a faster NPI	1.000	0.719	24	Resource management in global operations	1.000	0.622

5.1.1. Materials resource planning (MRP I)

Material requirements are calculated using bills of materials, inventory data, and the master production schedule. It makes recommendations on when to release material replenishment orders. Because it is time-phased, it also recommends rescheduling open orders when due dates and need dates are not in phase. Time-phased MRP begins with the items on the Master Production Schedule and determines the number of components and materials necessary to create those products, as well as the date the components and materials are needed. Exploding the bill of materials, adjusting for inventory quantities on hand or order, and offsetting the net requirements by the relevant lead periods results in time-phased MRP [8].

5.1.2. Manufacturing resource planning (MRP II)

MRP II (Manufacturing Resource Planning II) is a business-oriented information system. These systems are designed to collect, integrate, and process data in order to improve factory scheduling, design engineering, inventory management, and cost control. MRP II is a strategy for efficiently planning all of a manufacturing company's resources. It should include operational planning in units as well as financial planning in the relevant currency. The system must also have simulation capability to address "what if" questions. As seen in Fig. 1, it consists of numerous interrelated activities, including business planning, sales and operations (production planning), master production scheduling, materials requirements planning, capacity requirements planning, and capacity and material execution support systems. These systems' outputs also include financial reports such as a business strategy, purchase commitment report, shipping budget, and inventory projections [9].

5.1.3. Capacity requirements planning (CRP)

Capacity requirements planning refers to both a methodology and an enterprise application—software that manages a company's CRP process. Capacity Requirement Planning refers to the process of determining a company's production capacity and whether it can meet its production targets (CRP). A CRP analysis is an important management tool since it analyses whether a company's product can meet demand. Expenses of materials, routes, time requirements, lead-time, scheduled requests, and the condition of open orders too are required at each work centre. The level of detail is higher than in prior attempts. It is the only solution that necessitates the use of an ERP system [8].

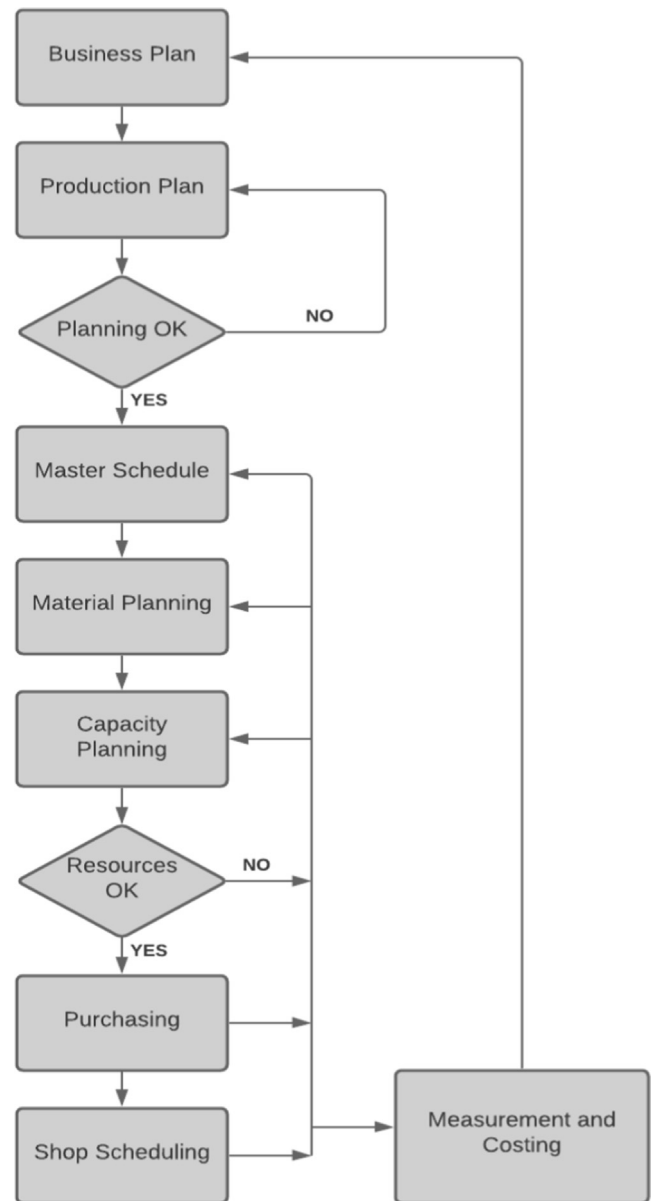


Fig. 1. Manufacturing Resource Planning II (MRP II) Process System [8].

5.1.4. Effect on manufacturing productivity using ERP systems

1. The average lead-time raw material inventory used by the organisation.

The greater the lead time, the more likely it is that changes to the order, including such raw - materials market prices or the danger of partial or complete cancellation, will be made. The longer the lead time, the more information need to monitor purchase orders. This variable represents the degree of uncertainty or complexity in the activity area of suppliers and raw - materials customer orders.

2. The Organization's Average Lead-Time Raw Material Inventory Remains.

By employing these solutions, manufacturers can save up to 30% on inventory holding costs (financing, storage, and insurance). The more the raw - materials inventories stay in the company, the larger the potential for cost savings. The variable demonstrates the impact of the change supported by the additional information on organisational goals related to material requirement planning [7].

3. The complexity of Product Specification.

The more intricate a product, the more data is required to manage consumer orders efficiently. This variable represents the level of ambiguity or complexity in the client order activity area. Three distinct sorts of items differ in their challenges inherent in tracking the purchase request.

- **Standard products:** Customers are unable to request product adjustments (e.g., food on store shelves). It's not difficult to track down consumer orders.
- **Modified versions of standard items:** Modifications are performed in response to consumer demands (for example, furniture that can be customized to meet the customer's needs). Hence, information on the particular specifications must be managed to ensure that the changes are implemented.
- **Custom-designed products:** Each order is a one-of-a-kind item made to the customer's specifications. The product's level of complexity is at its maximum. The company must develop a product-specific design, procure raw materials, and schedule.

4. The Number of Production Lines.

Two factors heavily influence the number of production lines: The company manufactures numerous goods, and then each item must go through multiple processes, including when different pieces are manufactured on separate manufacturing lines and then assembled. The more manufacturing lines there are, the more difficult it is to produce and coordinate. As a result, as the number of manufacturing lines increases, so does the amount of information needed to track raw material purchases and customer orders. The level of ambiguity or complication connected to customer order activity is denoted by this variable.

5. Customer Orders Have an Average Supply Time.

The longer the delay time for a customer order, the more complex the product. This is due to the fact that order specifications, natural resources, and other sources may differ. As a result, additional information is required to coordinate all operations during the order's duration. This variable represents the level of ambiguity or complexity in the client order venue [7].

5.2. Automation

5.2.1. Robotic process automation (RPA)

Robotic Process Automation (RPA) is one of the most widely explored areas by the industry due to its ability to transform traditional business processes on a variety of parameters. The software device substitutes human labour in repetitive tasks such as collect-

ing user data, performing operations on that data, scheduling operations and documenting the data or user actions. RPA may cover the gap between Cloud ERP and legacy applications, retaining the value of the traditional operation while adding cloud capabilities. It also overcomes the difficulties in the testing phase of enterprise systems by automating the testing process. Step 1: Identifying Processes to Automate: Developers must determine which operation they wish to automate, and run the procedure automatically for a certain data/user. RPA bots can understand the logic behind implementing such unique jobs and then perform them automatically and without interruption. Step 2: Mapping and Optimization of Processes: Creating a process plan is critical to ensuring that you understand the concept and each activity that must be taken to finish the process [14]. Fig. 2 depicts the life Cycle for RPA in ERP Systems.

The following are some examples of widely known jobs in firms that can use RPA in conjunction with their ERP systems to improve overall performance:

- **Inventory management:** Receiving automatic alerts when stock levels become too low or too high to maintain just the correct quantity.
- **Real-time Reporting generation:** This is one of the most important responsibilities because the RPA system can create real-time reports on multiple departments' process modifications. Also obtaining and transmitting sales, financial, or custom reports to the appropriate people as needed.
- **Customer relationship management:** Making accurate data and eliminating any manual work, keeping your data up to date. Also, with the usage of Chabot's, RPA automation can boost the efficiency of call centres, hence increasing customer satisfaction.
- **Human Capital Management:** In HCM, RPA can be utilized to automate all aspects of the employee experience, including recruiting, onboarding, and career development.
- **Finance:** By automating your Finance, users may eliminate costly errors in recurring financial and accounting operations such as invoice filing. RPA can also be configured to ensure compliance with regulations and legal reporting needs [15].

5.2.2. Artificial intelligence

The goal of artificial intelligence is to build software that functions like the human brain and can perform deep analysis in the same way that the human mind does. The demand for AI integration with ERP systems grows as it improves its capability to the point where it can analyse and generate decision/action suggestions using artificial neural networks. Artificial intelligence techniques with ERP are thought to be capable of discovering significant facts, patterns, relationships, and/or other sorts of new knowledge that would not have been discovered using traditional

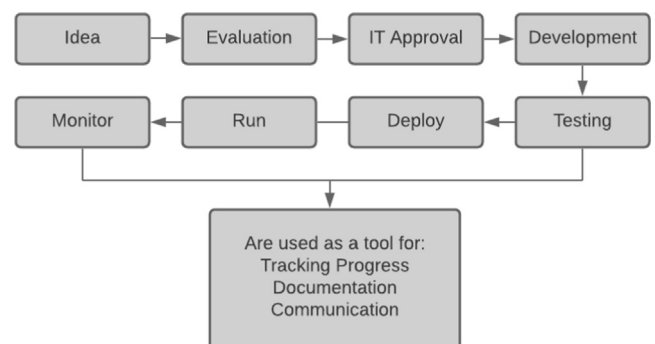


Fig. 2. Robotic Process Automation (RPA) life cycle for ERP systems [14].

analysis procedures. Implementation of AI with ERP is done by using different AI tools such as Machine Learning: Companies can profit from a machine learning-enabled ERP system in a variety of ways by using Predictive analytics, investigating the root causes and providing Optimization. Natural language interfaces/Virtual Agents: Operators with conversational AI skills can provide a more user-friendly interface to ERP systems that aren't always user-friendly [10]. Fig. 3 shows a model for Virtual Agents and Machine Learning for ERP systems.

Here are examples of how AI can influence ERP in various businesses:

- **Analytical Advances and Forecasting:** AI can work with enormous amounts of data, it can provide real-time, accurate data insights. It can analyze past data and forecast the future. These technologies detect periodic tendencies in the business and recommend whether you should reduce or boost production.
- **Inventory management:** Receiving automatic alerts when stock levels become too low or too high to maintain just the correct quantity.
- **Sales Automation:** Sales automation systems are concerned with tracking, reporting, and team productivity. The next phase of sales training will focus on making individual interactions sharper through the use of data. AI can be included early by bringing in vast amounts of data from outside the business to assist salespeople in developing thorough profiles of their targets.
- **Financial Management:** AI in financial management can automate repetitive accounting operations, boost transaction-processing speed, and verify the correctness of statements and reports.
- **Customer Service:** AI systems may learn from past customer service interactions, allowing chatbots to respond to enquiries more cost-effectively, rapidly, and accurately. Customer service quality can also be improved by combining real-time data from various customer-facing departments [11].

5.2.3. Industry 4.0

Industrialization is the framework that encompasses advancements in the production of materials under better conditions. Industry 4.0 seeks to reduce the complexity in production operations in order to increase efficiency and performance. By linking devices and equipment via the internet, the fourth industrial revolution enables more efficient manufacturing systems. These connections will strengthen ERP systems in areas such as customer interface, human-machine interface, partner interface, and staff interface through various means such as mobile apps. Integrating your corporate systems in compliance with Industry 4.0 regulations is becoming increasingly important [12]. The best way to accomplish this is to upgrade existing ERP systems with the most

recent technological integrations, as well as to adapt the ERP system to technical and process-related requirements. This includes data storage, data communication, and data utilization. How ERP integration with Industry 4.0 helps the companies:

- ERP solutions can use Big Data Technologies to collect and analyse data in real-time from numerous sources for decision making, quality control, and cost optimization. Advanced Analytics, in conjunction with using technology like wearable tech and virtual reality, has the potential to radically revolutionise maintenance services including logistic activities are handled.
- Independent manufacturing processes could be facilitated by using bots that can perform tasks intelligently, with a focus on safety, adaptability, variety, and cooperation, and by utilising integration approaches supported by ERP systems.
- The integration of Industry 4.0 innovation into ERP software will give businesses higher clarity into manufacturing processes and supply chain management for computer-controlled resource planning, allowing them to improve their logistical operations, increase inventory tracking, and better manage their personnel.
- Enterprises can combine big data analytics with AI algorithms for customer relationship management (CRM) and management of human resources (HRM) to process and analyse massive amounts of structured or unstructured customer data, identify hidden trends, and gain relevant insights into client data.
- The cloud enables organisations to efficiently handle mission-critical company information while increasing safety, scalability, and adaptability. It also allows corporate managers to easily access data from anywhere in the world and provides lower IT expenses, better workforce management, faster migration, and easier connection with other business tools [13].

5.2.4. Supply chain management

Business Intelligence complements traditional ERP capabilities. Department-specific software can be replaced with functionalities such as Process Management and Supply Chain Management (SCM). A supply chain is a dynamic process that comprises a continuous movement of information, materials, and finances between and among chain members. ERP systems are expected to increase SCM competencies in operational processes and improve customer responsiveness and satisfaction since they may automate company operations and enable process modifications. Supply chain improvements include forecast accuracy, delivery performance, fulfilment cycle time, inventory reduction, and overall productivity [16,19]. Supply chain management refers to the defined supply chain management among a network of inter-dependent manufacturers, suppliers, distribution centres, and retailers to improve the flow of goods, services, and information from original suppliers to final customers, intending to lower system-wide costs while maintaining required service standards [17].

Product inventories, stores, warehouses, and factory-floor waits have all been the subject of material flow management in the past. Managing such supplies entailed buying enough components ahead of time to ensure that long, uninterrupted manufacturing cycles were rarely interrupted by component shortages. By providing near real-time information on inventory levels, financing/payment, competitive costing, and forecasting requirements, ERP systems help supply chain relationships become more successful. According to the supply chain management council, optimizing these links/tasks in a company may save 6% of its total revenue. ERP for SCM focuses on improving the coordination of information flow along the distribution chain: demand, supply, production, delivery, and route optimization in the quest for a strategy to deliver the best product at the right location at the right time [18]. To achieve commercial success, it has been suggested that business

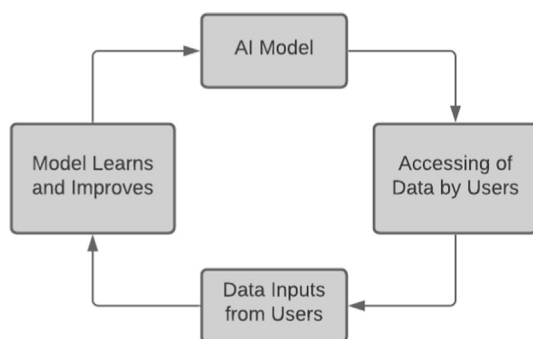


Fig. 3. Machine Learning and Virtual Agents model for ERP system [10].

process optimization throughout the value-added chain be complemented by current processing technology for corporation data processing optimization. Applications for market research and supply chain optimization can be connected with common ERP modules such as accounting, human resources, trading and marketing, and production. (20, 21).

Supply chain management activities include purchasing and supply management, operations management, marketing, logistics and transportation, and management information systems. Logistics researchers tend to concentrate on the transportation, distribution, and storage aspects of SCM, whereas operations management experts stress buyer–supplier collaboration and information exchange to reduce costs [29]. Barriers might include a lack of integration, management buy-in, measurement and availability of data challenges, and insufficient information systems. Shortening the product development life cycle, reducing inventory, lowering manufacturing costs, improving quality and increasing on-time order fulfilment, are just a few of the concrete benefits of ERP on SCM adoption. Indirect benefits include improved service quality and faster response times in consumer needs, information sharing and exchange, and continuous provision of accurate, timely, and relevant information. Other ways ERP can help improve SCM performance include internal business process integration, improved information flow between departments, improved partnerships and collaboration with outsourcing suppliers and customers, global sourcing, and client service flexibility [22,24].

6. Conclusion

This comprehensive review of the literature has provided insights into common crucial success criteria and ERP system deployment in a variety of industries. ERP system and a company's business and manufacturing operational productivities are inextricably linked and appropriate strategy and solutions can be implemented to achieve an even higher level of these productivities. Certain post-implementation activities, such as software updating and maintenance, are critical in the lifecycle of an ERP system in order to maximise the system's economic benefits. Furthermore, business knowledge, practice, and procedures may differ between industries which is why an increased effort is required to identify the best practices for each organization. The efficiency of processes of manufacturing management especially those involving Industry 4.0, Robotics and Automation are highly fuelled by ERP systems, as they provide modern solutions and statistical data of respective processes to measure and increase profitability. In the future, ERP systems will have direct exposure to all departmental data. Consequently, they have all the information needed to forecast market trends. ERP users can look forward to more robust data discovery capabilities and an all-in-one tool capable of processing both structured and unstructured data. ERP trends will start to evolve as well as develop, as these systems involving Modern Manufacturing Management play an integral role in increasing the productivity of the various industries. As per the survey results referred to and presented above, an average of 63% increase in efficiency was observed across factors of commonalities, keeping in mind the system pre-design requirements planning and implementation to be key factors in the success of an ERP program for an organization. ERP, at its core, allows employees to do their jobs more efficiently by reducing obstacles between departments. ERP success gives a global, continuous view of data that may allow firms to address issues effectively and accelerate progress.

CRedit authorship contribution statement

R. Chopra: Resources, Supervision, Writing – review & editing.
L. Sawant: Resources. **D. Kodi:** Resources. **R. Terkar:** Supervision, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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